



Inspiring Excellence

Introduction to Computer Network Ethics

Lecture 0.3 | CSE421 – Computer Networks

Department of Computer Science and Engineering
School of Data & Science

Objectives

- Understand what **morality** and **ethics** mean
- Learn why **ethics matter in computer networks**
- Understand **responsible behavior online**

What is Morality

- Morality means **knowing right and wrong**
- It guides how people **should behave**
- Morality depends on **culture, place, and time**
- What is right in one society may be wrong in another

What is Ethics

- Ethics is the **study of right and wrong actions**
- It helps us **decide what is fair and correct**
- Ethics builds **trust and cooperation** in society
- It helps justify **why an action is right or wrong**



Morality vs Ethics

Example :

Morality: You feel it is wrong to read someone else's email.

Ethics: University IT policy clearly states that accessing others' emails is not allowed.

Ethical Theories

Ethical theories help us judge actions.

Two main types:

- **Consequentialist:**

- Looks at the **result** of an action
- Example: “Did this action help more people?”

- **Deontological:**

- Looks at the **intention and rules**
- Example: “Was this action right, even if the result was bad?”

Examples

Consequentialist Ethics

- A student shares lecture notes with the whole class before an exam.
- This helps many students prepare better.

Thinking (Consequentialist view):

The action is ethical because the **result helps more people.**

Examples

Deontological Ethics

A student finds the exam question paper before the exam. Even if sharing it could help friends score better, the student does not share it.

Thinking (Deontological view):

The action is ethical because **sharing is wrong**, even if the result might help others.

Ethical Principles

An action is ethical if:

1. It **benefits society**
2. It **respects people** and does not use them
3. It is **fair to everyone**

Professional Codes of Ethics

- A code of ethics tells professionals **how to behave**
- Helps maintain **high standards and trust**
- Guides ethical decision-making
- Example: **ACM Code of Ethics** for computing professionals
 - Association for Computing Machinery's Code of Ethics can be found here: www.acm.org/about/code-of-ethics

Computer Ethics

Computer ethics focuses on:

- **Privacy** – protect personal data
- **Accuracy** – keep data correct
- **Property** – respect ownership of software and data
- **Access** – control who can see or use information

Computer Ethics for Computer Professionals

Professionals should show:

- **Competence** – work only in areas they understand
- **Responsibility** – protect confidential information
- **Integrity** – be honest and fair

Why Ethics Matter in Networking

- Unethical actions can cause **serious harm**
- Example:
 - Sharing a password
 - Private data gets leaked
 - Trust is broken
- Ethical behavior keeps networks **safe and reliable**
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Think

- A student shares university Wi-Fi or LMS access with outsiders.
- During exams, the network becomes slow for everyone.

Think About This:

Is it ethical to share access if it harms other students?



WHERE do ethical problems actually happen?

Internet Privacy

- Emails are **not fully private**
- Copies of emails are stored on many computers
- Websites use **cookies** to track activity
- **Spyware** secretly collects user information

Think

- Many people use shared computers in labs, offices, or cyber cafés.
- Forgetting to log out can expose personal email or social media.

Think About This:

If you see someone else's email already open, what should you do?

Computer Crime

Computer crime means **using computers for illegal acts**

Common criminals:

- Employees (most common)
- Hackers and crackers
- Organized crime groups

Types of Exploits

- **Virus** – spreads with user action
- **Worm** – spreads automatically
- **Trojan Horse** – looks safe but is harmful
- Others: phishing, DDoS, spam, ransomware

Think

- Fake “Govt job circular” or “scholarship” emails with links.
- Malware spread through USB drives used in computer labs.

Think About This:

How can one careless click affect many users?

How to Stay Ethical Online

- Use networks only for **study and work**
- Protect **privacy and passwords**
- Avoid malware and phishing
- Respect **copyright and licenses**
- Report problems quickly

Think

- Downloading movies or software using campus internet.
- Using cracked software to finish assignments.

Think About This:

Is something ethical just because “everyone does it”?

Password Safety

- Use **strong passwords**
- Never share passwords
- Change passwords if compromised
- Weak passwords can lead to **serious consequences**
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Think

- Sharing LMS or Google account password with a friend.
- Friend submits wrong or late work using your account.

Think About This:

Who should be responsible if your password is misused?

Reporting Incidents

- Report security problems immediately
- Reporting is **not complaining**
- It helps protect **everyone**

Think

- Reporting fake exam result links or phishing emails.
- Informing IT about suspicious Wi-Fi or login activity.

Think About This:

What happens if no one reports a security issue?

Summary & Key Takeaways

- Apply Ethics online and offline
- Respect privacy and data
- Follow rules and ask when unsure
- Ethical behavior builds **trust and reputation**

Reading

- Books:

1. “Computer Network Security and Cyber Ethics” by Joseph Migga Kizza
 - Focused on chapter 2 & 3
2. “Ethics in Information Technology” by George Reynolds
 - Focused on chapter 2 & 3